



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	PROPERTIES OF SOLIDS				
Type of Course	Major Elective (SPECIALIZATION I: MATERIALS SCIENCE)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	The prerequisites for the course on crystal structure, theory of solids, semiconductor properties, and dielectric and magnetic properties of solids include a solid foundation in physics, mathematics, quantum mechanics, chemistry, electricity, and magnetism.				
Course Summary	The course provides a comprehensive study of crystal structure, theory of solids, semiconductor properties, and dielectric and magnetic properties of solids, aiming to understand the fundamental principles governing the behavior of materials in these domains.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate an understanding of crystal structures and their impact on material properties	Apply	Conceptual Understanding	Examinations, Assignments
CO2	Analyze the theoretical models of solids and their applicability to real-world scenarios	Analyze	Application	Problem Sets, Case Studies

CO3	Evaluate semiconductor properties and their role in electronic device functionality	Evaluate	Application	Laboratory Experiments, Projects
CO4	Explain the principles underlying dielectric properties of solids and their technological applications	Understand	Conceptual Understanding	Presentations, Written Reports
CO5	Investigate magnetic properties of solids and their implications in magnetic storage and sensing technologies	Evaluate	Application	Research Papers, Presentations
CO6	Synthesize knowledge of crystal structure, theory of solids, semiconductor, dielectric, and magnetic properties to propose solutions to complex material-related problems	Create	Synthesis	Capstone Projects, Oral Defenses
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 + 12)	Marks (70)
I	CRYSTAL STRUCTURE		15	20
	1	Crystal lattice and translation vectors, unit cell, basis	3	
	2	Symmetry operations, point groups and space groups	3	
	3	Types of lattices, lattice directions and planes, inter planar spacing	3	
	4	Simple crystal structures with examples	3	
	5	X-ray diffraction and reciprocal lattice. Brillouin zones	3	

	Sections 1.1 – 1.13, 2.1 – 2.2, 2.4, 2.7 of Book 1			
II	THEORY OF SOLIDS		10	15
	6	Drude – Lorentz’s classical theory	1	
	7	Sommerfeld’s quantum theory- Free electron gas in one dimension	2	
	8	Fermi energy, Total energy Density of states, Filling of energy levels	2	
	9	Application of free electron gas model	1	
	10	Band Theory of solids-Bloch theorem-Kronig Penney model-velocity and effective mass of electron	3	
	11	Distinction between metal, insulator and semiconductors	1	
	Sections 5.1 – 5.3, 6.1 – 6.5 of Book 1			
III	SEMICONDUCTOR PROPERTIES		10	10
	12	Semiconductors – Intrinsic and Extrinsic	2	
	13	Drift velocity	2	
	14	Mobility and conductivity of Intrinsic semiconductors	2	
	15	Carrier concentration, Fermi level	2	
	16	Conductivity for intrinsic and extrinsic semiconductors	2	
	Sections 7.1 – 7.6 of Book 1			
IV	DIELECTRIC AND MAGNETIC PROPERTIES OF SOLIDS		13	25
	17	Types of Magnetism – origin of permanent magnetic moment	1	
	18	Diamagnetism and Paramagnetism (classical theory), ferromagnetism (Weiss theory)	4	
	19	Antiferromagnetism and ferrimagnetism (Qualitative ideas only)	2	
	20	Polarisation, Susceptibility, Local field	2	

	21	Dielectric constant and polarizability and its sources	2	
	22	Ferro and Piezo electricity (Qualitative ideas only)	2	
	Sections 8.1 – 8.7, 9.1 – 9.7 of Book 1			
V	OPEN ENDED MODULE: CRYSTAL BONDING AND DEFECTS IN CRYSTALS		12	

Books and References:

- 1.Solid State Physics by R. K. Puri and V. K. Babbar (Book 1)
- 2.Solid State Physics by S O Pillai 6th Edition (Book 2)
- 3.Solid State Physics: Structure and Properties of Materials by M. A. Wahab (Book 3)

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	2	2	0	0	0	0	0	0	0	0	0
CO 2	2	2	2	2	2	0	0	0	0	0	0	0	0
CO 3	2	3	2	2	2	0	0	0	0	0	0	0	0
CO 4	2	2	2	2	0	0	0	0	0	0	0	0	0
CO 5	2	3	2	2	2	0	0	0	0	0	0	0	0
CO 6	3	3	2	2	2	3	3	2	0	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓